PROJECT PROGRESS REPORT: SMARTGROW

Project Name: SmartGrow – Smart Irrigation System	**Reporting Period:** Day 1 to 7
Team Name: Team Yantra (Priyalal S.D., Jayaweera W.P.K., Gunathilaka S.N., Lochana H.M.V.H.K.)	**Track:** Track A – Embedded IoT System Development

1. Project Overview

SmartGrow is an automatic plant watering system. Instead of watering plants on a fixed timer, it measures actual soil moisture using a sensor. An ESP32 microcontroller turns the water pump ON when the soil is dry and turns it OFF when the soil is wet. Users can also view moisture levels and irrigation history using a mobile app.

2. How the System Works

SOIL SENSOR (Read Moisture) --> ESP32 (Check Thresholds) --> PUMP DRIVER (Water Soil) --> MOBILE APP (Show Status & History)

3. Project Progress (10-Week Plan)

Day	Task Focus	Status	Summary of Completed Work
Day 1-3	System Design	**Completed**	Selected all hardware parts and planned system diagram.
Day 4-5	Sensor Testing	**Completed**	Tested moisture sensor and calibrated dry/wet readings.
Day 6-	Pump Control	**In Progress**	Building motor driver circuit and added auto ON/OFF control logic.
	Hardware Build	**Not yet**	----------.
	Testing & Testing	**Not yet**	----------.
	Final Prep	**Not yet**	----------.

4. Key Results and Safety Features

- **Accurate Sensing:** Sensor readings were calibrated using dry and wet soil samples for reliable results.
- **Smooth Pump Control:** Used two set points (low and high limits) to stop the pump from turning ON and OFF too fast.
- **Safety Timer:** Set a maximum pump run time to prevent flooding or dry pumping if water runs out.
- **Offline Working:** The system waters plants locally on the ESP32 even if the Wi-Fi connection drops.

5. Next Steps

1. Perform final trial run of the complete system demo.
2. Finalize recorded water-saving data for competition submission.